University of Michigan

NERS 515

Nuclear Measurements Laboratory

Laboratory 4: Geiger-Müller Counting

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Abstract

1 Introduction

Geiger-Müller (G-M) tubes are one of the oldest radiation detectors and are still widely used because they are simple, cheap and easy to operate (Knoll, p. 207) [1]. The strong electric field in a G-M tube makes every event grow into a chain of avalanches of the same size, so the pulses are large, of the order of volts, and the tube needs little or no external amplification. When only a simple, economical counting system is needed, a G-M tube is often the best choice (Knoll, p. 207). The trade-offs are that the pulse height carries no energy information, so the tube can only count events, and that its dead time is longer than that of any other commonly used detector, which limits it to low count rates (Knoll, pp. 207–208). In this laboratory we characterize a G-M tube as a counting system. We find its operating voltage from the counting curve and the pulse height as the high voltage is raised, compare the pulse heights from three beta emitters, estimate the absorption coefficient and endpoint energy of a beta source with aluminum absorbers, and measure the dead time with an oscilloscope and with the two-source method [2].

2 Theory

PARTNER: Theory. Equations used in the calculations below are listed with their sources in the handoff list (05_report_outline.md, section D).

3 Equipment and Procedure

3.1 Equipment

Details of all equipment used in this lab are listed in Table 1. The list does not include the serial numbers of the SCA and the High Voltage Power Supply because we were advised not to remove the instrument from the NIM, in case it caused other issues. In addition, we used a G-M tube, a preamplifier, a multichannel analyzer, and Th-232, Co-60, C-14 and Cl-36 sources as specified in the lab manual [2]. Figure 1 shows the connections.

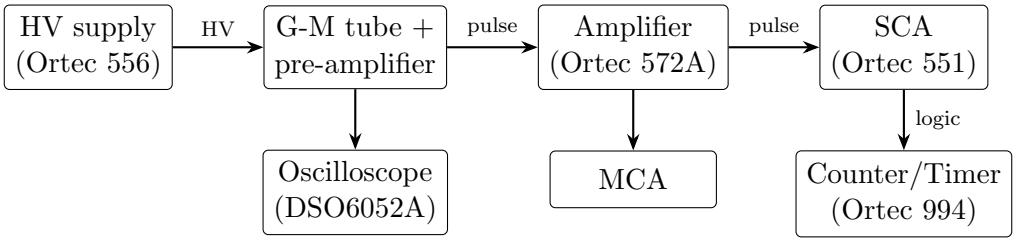


Figure 1: Block diagram of the setup.

Purpose	Manufacturer	Model	Serial Number
Oscilloscope	Agilent Technologies	DSO6052A	MY44007040
High Voltage Power Supply	Ortec	556	-
Amplifier	Ortec	572A	121
Single-Channel Analyzer	Ortec	551	-
Counter/Timer	Ortec	994	2108

Table 1: List of equipment used in Laboratory 4.

3.2 Experimental Procedure

We followed the lab manual [2] for all four measurements. The settings we had to find or choose are listed in Table 2.

Counting curve and pulse height vs. HV. With a source on the tube, we raised the high voltage from 650 V to 910 V in 20 V steps. At each step we waited 30 s for the voltage to settle, counted for 3 s with the counter/timer, and read the pre-amplifier pulse height on the oscilloscope. The pulse height uncertainty is the half width of the band of pulses, and the baseline correction is already applied to the tabulated values. We took 840 V, on the high side of the plateau, as the operating voltage. The noise LLD (0.1 V) was found at 800 V with the source removed.

Pulse height consistency. At 840 V we recorded the pulse height spectra of Co-60, C-14 and Cl-36 with the MCA for 30 s real time each and took the centroid channel of the peak. We converted channels to volts with the calibration from our Lab 2 report [3].

Beta attenuation. We placed Al absorbers of increasing mass thickness between the Co-60 source and the tube, and counted for 5 to 14 s at each thickness (longer for thicker absorbers). The zero-absorber row in Table 5 is a background count.

Dead time. We measured the dead time and recovery time on the oscilloscope with the cursors on the persisting trace of the pulses, then used the two-source method with Th-232 and Cl-36. The sources sat on the 3rd shelf, about 3 cm from the tube and about 3 cm apart. A 1 mm change in position changes the rate noticeably, so the source that stays in place was never moved; only the other was inserted and removed. Each of the three runs has four 10 s counts: background, each source, and both together.

4 Data

4.1 Counting curve and pulse height vs. high voltage

Table 3 lists the raw counts (3 s each) and the pre-amplifier pulse heights with their uncertainties. Counts first appeared at 670 V (44 counts), but no pulse was visible on the scope, so we attribute them to noise. Pulses first showed up on the scope at 690 V. At 910 V the counts jumped to 267 (177 at 890 V) and we stopped the scan.

Setting	Value
High voltage scan	650 to 910 V, 20 V steps, 30 s settle, 3 s count
Operating voltage	840 V
Noise LLD	0.1 V (found at 800 V, source removed)
SCA mode	integral
Scope baseline correction	-2.625 mV
Scope for dead time	100 μs/div, Normal trigger, ∞ persist, trigger level -78.2 mV
MCA spectra	30 s real time
Channel to voltage (Lab 2 [3])	$V = -36.0\ \text{mV} + (1.31\ \text{mV})\, n$, with n the channel number
Two-source counts	10 s each, 3rd shelf, sources about 3 cm from the tube

Table 2: Settings we found or chose.

HV (V)	Counts (3 s)	Pulse height (mV)	HV (V)	Counts (3 s)	Pulse height (mV)
650	0	0 ± 0	790	169	137.5 ± 28.75
670	44	0 ± 0	810	178	155.31 ± 29.06
690	119	18.75 ± 2.75	830	183	212.81 ± 19.06
710	152	31.125 ± 4.13	850	185	255 ± 21.25
730	163	47.88 ± 9.13	870	178	303.75 ± 35
750	164	70.5 ± 10.75	890	177	363.75 ± 52.5
770	169	97.63 ± 14.38	910	267	443.75 ± 56.25

Table 3: Counts in 3 s and pre-amplifier pulse height $\pm$ half width versus high voltage.

4.2 Pulse height consistency

Table 4 gives the MCA centroid channel and gross counts for each source.

Source	Centroid channel	Gross counts	Real time (s)
Co-60	2906	4419	30
C-14	3374	1280	30
Cl-36	1939	13986	30

Table 4: MCA centroid channel and gross counts for each source at 840 V.

CELING: add the MCA spectra of Co-60, C-14 and Cl-36 here as Figure 2 (placeholder).

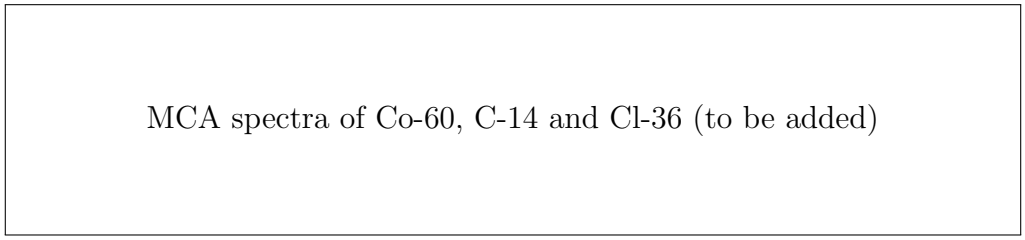


Figure 2: MCA pulse height spectra of the three sources at 840 V, 30 s real time.

4.3 Beta attenuation

Table 5 lists the absorber thickness, counts and counting time. The thickness uncertainty is 10% of the thickness [2].

Absorber	Mass thickness (mg/cm^2)	Counts	Time (s)
Background (no source)	-	5	5
Al	14.67	340	5
Al	66.53	83	5
Al	177.22	110	8
Al	415.80	112	8
Al	628.78	94	8
Al	843.34	139	12
Al	1278.61	84	14
Al	1540.52	100	14

Table 5: Counts and counting time for each aluminum absorber with the Co-60 source.

4.4 Dead time

Figure 3 shows the oscilloscope traces with the cursors placed as described in the manual [2]. The dead time runs from the initial pulse (-2 μs) to where pulses start to appear again (358

μs); the second measurement runs from there to where the pulses reach full height again (838 μs). The cursor step is 2 μs.

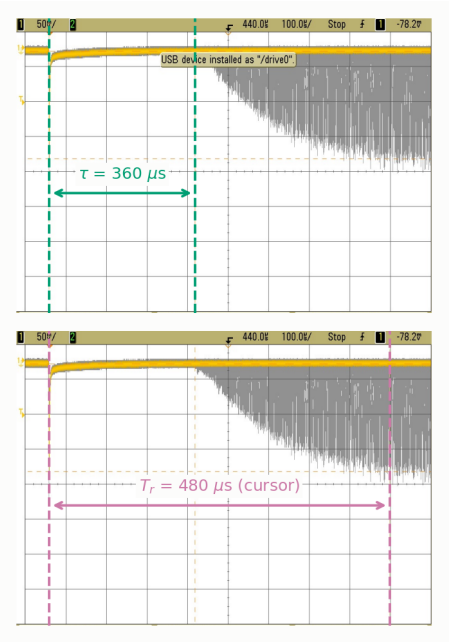


Figure 3: Oscilloscope traces with cursors: dead time (top, 360 μs) and recovery measurement (bottom, 480 μs).

Table 6 has the two-source counts.

Run	Background	Th-232	Cl-36	Both
1	5	1174	446	1585
2	2	722	484	1163
3	1	1056	441	1412

Table 6: Two-source counts for each run (10 s per count).

5 Analysis of Data

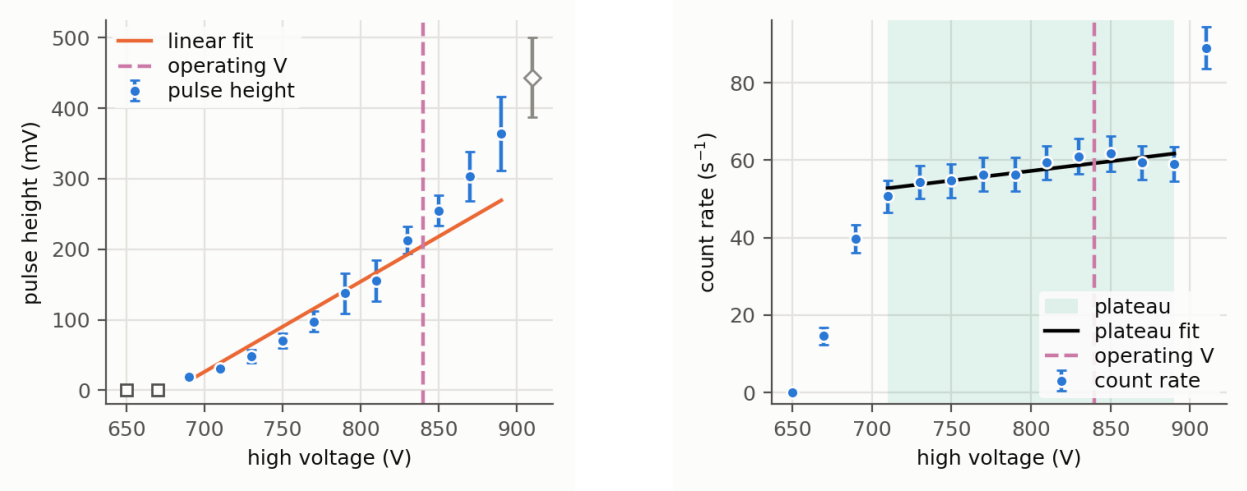
Count uncertainties are Poisson, $\sigma_N = \sqrt{N}$, propagated with Knoll Eq. 3.37 and its special cases (Eqs. 3.38–3.42) [1]. A rate from N counts in time t is $r = N/t$ with $\sigma_r = \sqrt{N}/t$ (Knoll, Eqs. 3.39–3.40).

5.1 Counting curve and pulse height vs. high voltage

Count rates. At 830 V we counted 183 in 3 s, so $r = 183/3 = 61.0$ s^{-1} and $\sigma_r = \sqrt{183}/3 = 4.5$ s^{-1}. All rates are in Table 7 and Figure 4b. The rates at the two voltages around the 840 V operating point are 61.0 ± 4.5 s^{-1} (830 V) and 61.7 ± 4.5 s^{-1} (850 V).

HV (V)	Rate (s^{-1})	HV (V)	Rate (s^{-1})
650	0.0 ± 0.0	790	56.3 ± 4.3
670	14.7 ± 2.2	810	59.3 ± 4.4
690	39.7 ± 3.6	830	61.0 ± 4.5
710	50.7 ± 4.1	850	61.7 ± 4.5
730	54.3 ± 4.3	870	59.3 ± 4.4
750	54.7 ± 4.3	890	59.0 ± 4.4
770	56.3 ± 4.3	910	89.0 ± 5.4

Table 7: Count rate $r = N/3$ s versus high voltage.



(a) Pulse height

(b) Counting curve

Figure 4: Pulse height (a) and count rate (b) versus high voltage. Dashed line: 840 V operating voltage. Error bars are the half width of the pulse band in (a) and $\sqrt{N}/t$ in (b).

Plateau slope. Knoll quotes the plateau slope in percent per 100 V (p. 215) [1]. We take the plateau as 730–890 V and compare the mean rate of the three lowest points (730, 750, 770 V) with that of the three highest (850, 870, 890 V). A group rate is the summed counts over the summed time, with $\sigma = \sqrt{\sum N}/\sum t$ (Knoll, Eq. 3.42). Then

$$s = \frac{r_2 - r_1}{V_2 - V_1}, \qquad \sigma_s = \frac{\sqrt{\sigma_{r_1}^2 + \sigma_{r_2}^2}}{V_2 - V_1}, \qquad s_{\%} = 100\,\frac{s\,(100\text{ V})}{r_1}, \tag{1}$$

using Knoll Eq. 3.38 for σ_s and the rate at the start of the plateau, r_1, as the reference rate. Sample calculation: $r_1 = 496/9$ s $= 55.1 \pm 2.5$ s^{-1} at $V_1 = 750$ V and $r_2 = 540/9$ s $= 60.0 \pm 2.6$ s^{-1} at $V_2 = 870$ V, so

$$s = \frac{60.0 - 55.1}{120\text{ V}} = 0.041 \pm 0.030\text{ s}^{-1}\text{V}^{-1}, \qquad s_{\%} = 7.4 \pm 5.4\ \%\text{ per 100 V}.$$

PARTNER: discuss the counting curve and pulse height vs. high voltage.

5.2 Pulse height consistency

The Lab 2 calibration is channel $= 765.8$ V $+ 27.6$ [3], which we use as $V = -36.0$ mV $+$ (1.31 mV)$\,n$, with the Lab 2 oscilloscope amplitude uncertainty of 25.4 mV [3]. For Co-60, $V = -36.0$ mV $+ 1.31$ mV $\times 2906 = 3.77$ V. The rate is $r = N/30$ s with $\sigma_r = \sqrt{N}/30$ s, so for Co-60 $r = 4419/30 = 147.3 \pm 2.2$ s^{-1}. Results are in Table 8 and Figure 5.

Source	Amplitude (V)	Rate (s^{-1})
Co-60	3.77 ± 0.03	147 ± 2
C-14	4.38 ± 0.03	43 ± 1
Cl-36	2.50 ± 0.03	466 ± 4

Table 8: Pulse amplitude (MCA centroid) and count rate at 840 V.

PARTNER: discuss the pulse height consistency.

5.3 Beta attenuation

The net rate is the gross rate minus the background rate (Knoll, Eqs. 3.52–3.53) [1],

$$r = \frac{G}{t_g} - \frac{B}{t_b}, \qquad \sigma_r = \sqrt{\frac{G}{t_g^2} + \frac{B}{t_b^2}}, \tag{2}$$

with G, t_g the gross counts and time and B, t_b the background counts and time. The background is $B/t_b = 5/5 = 1.0 \pm 0.45$ s^{-1}. For 14.67 mg/cm^2: $G/t_g = 340/5 = 68.0 \pm 3.7$ s^{-1}, so $r = 68.0 - 1.0 = 67.0 \pm 3.7$ s^{-1} and $\ln r = 4.20 \pm 0.055$, using $\sigma_{\ln r} = \sigma_r/r$ (Knoll, Eq. 3.37). The thickness uncertainty is 10% of the thickness [2]. Table 9 and Figure 6 show the results.

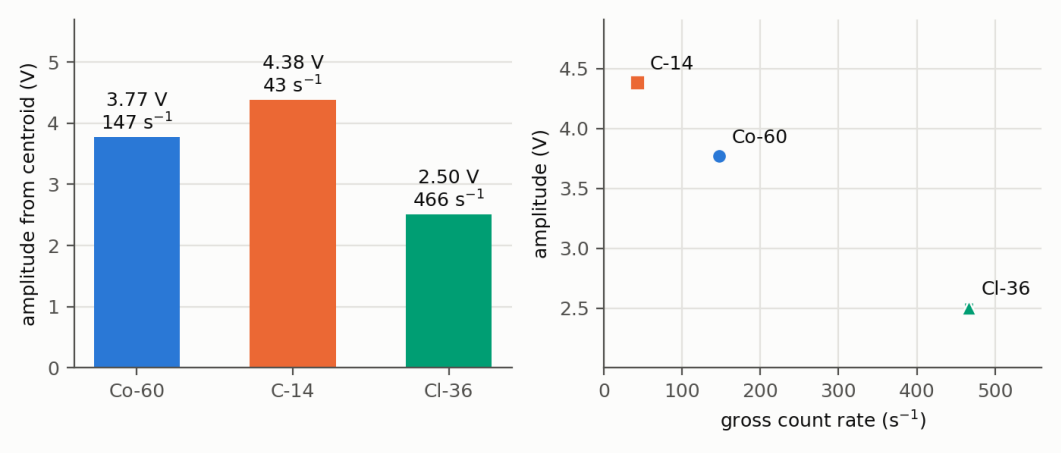


Figure 5: Pulse amplitude from the MCA centroid and count rate for each source.

Thickness (mg/cm^2)	**Gross rate (s^{-1})**	**Net rate (s^{-1})**	$\ln r$
14.7	68.00 ± 3.69	67.0 ± 3.7	4.20 ± 0.06
66.5	16.60 ± 1.82	15.6 ± 1.9	2.75 ± 0.12
177.2	13.75 ± 1.31	12.8 ± 1.4	2.55 ± 0.11
415.8	14.00 ± 1.32	13.0 ± 1.4	2.56 ± 0.11
628.8	11.75 ± 1.21	10.8 ± 1.3	2.37 ± 0.12
843.3	11.58 ± 0.98	10.6 ± 1.1	2.36 ± 0.10
1278.6	6.00 ± 0.65	5.0 ± 0.8	1.61 ± 0.16
1540.5	7.14 ± 0.71	6.1 ± 0.8	1.82 ± 0.14

Table 9: Gross rate, net rate and $\ln r$ for each absorber.

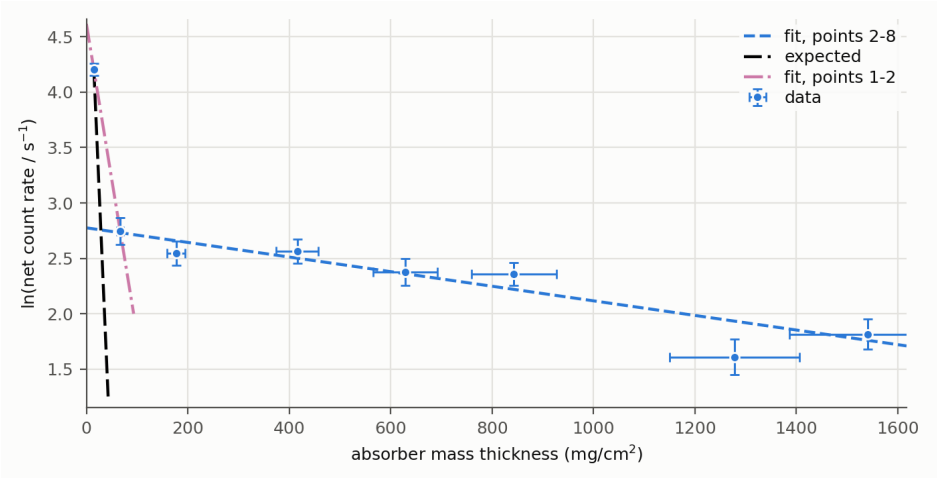


Figure 6: Natural log of the net count rate versus Al mass thickness. Dash-dot line: through points 1 and 2; dashed line: fit to points 2–8; black dashed line: expected n.

The beta count rate falls approximately as $I/I_0 = e^{-nt}$ with t the mass thickness (Knoll, Eq. 2.14, p. 45) [1]. We take n from the first two points, since the rest of the data is nearly flat (Figure 6):

$$n = \frac{\ln(r_1/r_2)}{t_2 - t_1}. \tag{3}$$

The error follows from Knoll Eq. 3.37 with σ_{r_1}, σ_{r_2} and the 10% thickness uncertainties as independent inputs. With $\ln(67.0/15.6) = 1.457$ and $t_2 - t_1 = 0.05186$ g/cm^2,

$$n = 28.1 \pm 4.5 \text{ cm}^2/\text{g}.$$

We read the expected value for the Co-60 beta from Knoll Fig. 2.16 [1]: $n_{\text{exp}} \approx 105 \pm 10$ cm^2/g at the marked Co-60 endpoint energy (≈ 0.32 MeV). For the endpoint energy from our n we inverted the line of Fig. 2.16, approximated by $E = 1.47\,(n/10)^{-0.668}$ MeV (our own reading of the log-log figure, not an equation from Knoll). This gives $E_{\max} = 0.74 \pm 0.08$ MeV (error from σ_n only), and 0.31 MeV for n_{exp}.

PARTNER: discuss the beta attenuation (comparison with the expected n, endpoint energy).

5.4 Dead time

Oscilloscope. The cursors in Figure 3 give a dead time of 360 ± 3 μs (-2 to 358 μs). The uncertainty is one cursor step (2 μs) on each cursor, added in quadrature. The interval from the end of the dead time to full pulse height is 480 ± 3 μs (358 to 838 μs), and the recovery time measured from the initial pulse is 840 ± 3 μs (-2 to 838 μs).

Two-source method. With background rate m_b, source rates m_1 (Th-232) and m_2 (Cl-36) and combined rate m_{12}, each the 10 s count divided by 10 s, the dead time of a non-paralyzable system is (Knoll, Eqs. 4.30–4.32) [1]

$$X = m_1 m_2 - m_b m_{12}, \qquad Y = m_1 m_2 (m_{12} + m_b) - m_b m_{12}(m_1 + m_2), \tag{4}$$

$$Z = \frac{Y\,(m_1 + m_2 - m_{12} - m_b)}{X^2}, \qquad \tau = \frac{X\left(1 - \sqrt{1 - Z}\right)}{Y}. \tag{5}$$

The uncertainty of τ is propagated with Knoll Eq. 3.37 from $\sigma_N = \sqrt{N}$ on the four counts. For run 1: $m_b = 0.5$, $m_1 = 117.4$, $m_2 = 44.6$ and $m_{12} = 158.5$ s^{-1} give $X = 5157$ s^{-2}, $Y = 8.20 \times 10^5$ s^{-3}, $Z = 0.092$, and $\tau = 298 \pm 566$ μs. The mean of the three runs and its standard error $s/\sqrt{3}$, with s the sample standard deviation (Knoll, Eqs. 3.5, 3.9 and 3.37), are $\bar{\tau} = 627 \pm 195$ μs ($s = 337$ μs). Table 10 and Figure 7 compare the runs with the oscilloscope dead time.

PARTNER: discuss the two dead-time measurements (which to trust, the model assumed, recovery time).

6 Conclusions

PARTNER writes this section.

Run	τ (μs)
1	298 ± 566
2	612 ± 729
3	971 ± 634
Mean ± SEM	627 ± 195

Table 10: Two-source dead time per run; SEM is the standard error of the mean.

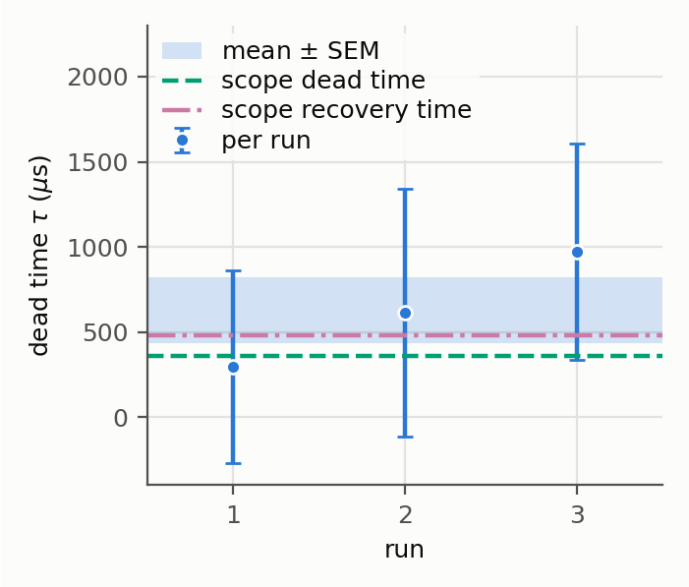


Figure 7: Two-source dead time per run, mean ± SEM, and the scope dead and recovery times.

References

[1] G. F. Knoll, *Radiation Detection and Measurement*. Hoboken, NJ: John Wiley & Sons, Inc., 4th ed., 2010.

[2] University of Michigan, Department of Nuclear Engineering and Radiological Sciences, "NERS 515 nuclear measurements laboratory: Laboratory 4," 2026.

[3] C. Li and C. A. Korfiatis, "NERS 515 laboratory 2: Single-channel and multichannel pulse analysis," 2026. Lab report, University of Michigan.